

Wazuh SIEM Home Lab Portfolio Report

Ubuntu Server 24.04 • Wazuh 4.14 • Windows Endpoint • Sysmon

Executive Summary

This project documents the deployment of a Wazuh SIEM home lab built in Oracle VirtualBox. The environment included a Wazuh server running on Ubuntu Server 24.04 and a Windows endpoint integrated through the Wazuh Agent. Sysmon telemetry and endpoint events were successfully collected and viewed through Wazuh Threat Hunting.

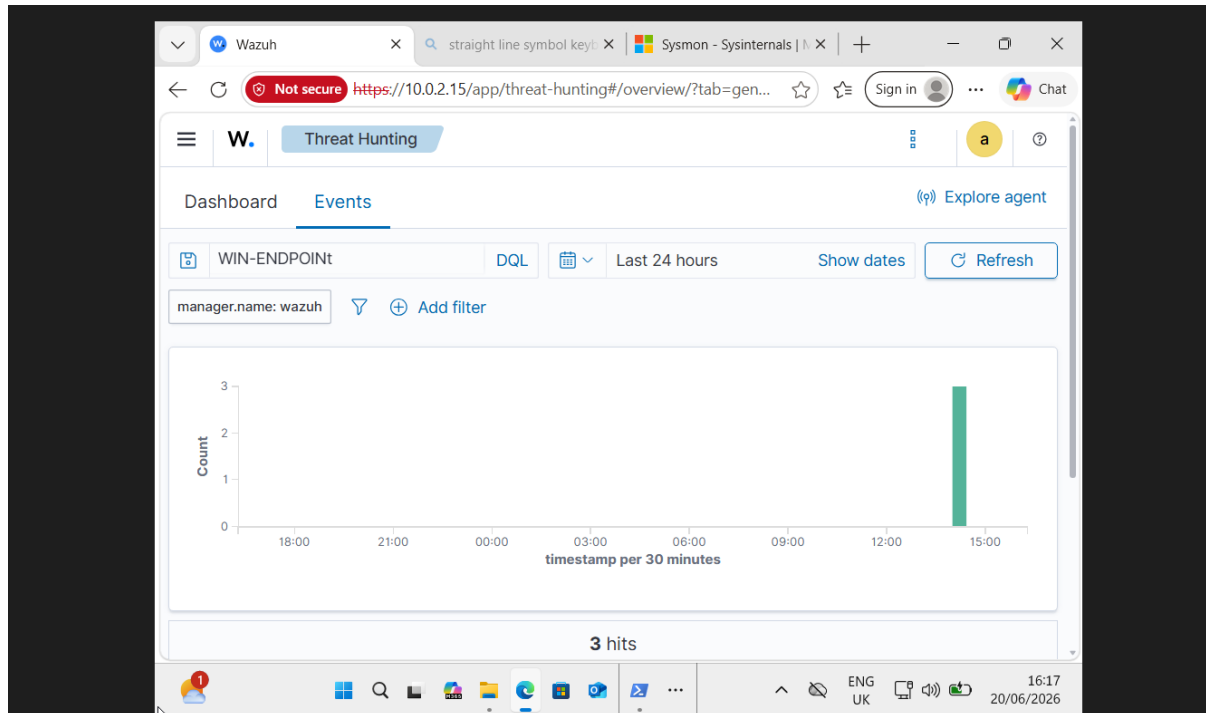
Technologies Used

- Wazuh 4.14.x
- Ubuntu Server 24.04 LTS
- Windows Endpoint (Windows 11)
- Sysmon
- Oracle VirtualBox
- PowerShell

Project Achievements

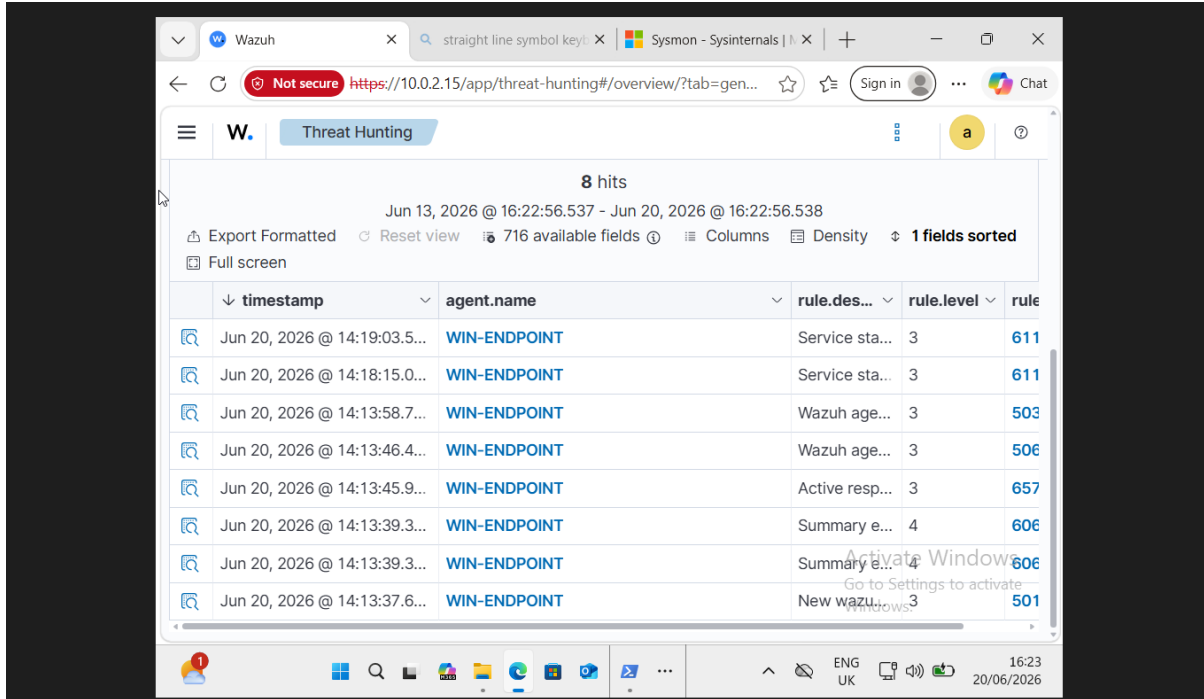
- ✓ Installed and configured Wazuh Manager, Indexer, Filebeat and Dashboard
- ✓ Resolved Ubuntu compatibility issues
- ✓ Corrected root partition sizing issues
- ✓ Configured NAT port forwarding for dashboard access
- ✓ Onboarded a Windows endpoint to Wazuh
- ✓ Validated endpoint telemetry and agent connectivity
- ✓ Reviewed events through Threat Hunting

Threat Hunting Dashboard



Threat Hunting view showing telemetry from WIN-ENDPOINT.

Endpoint Events

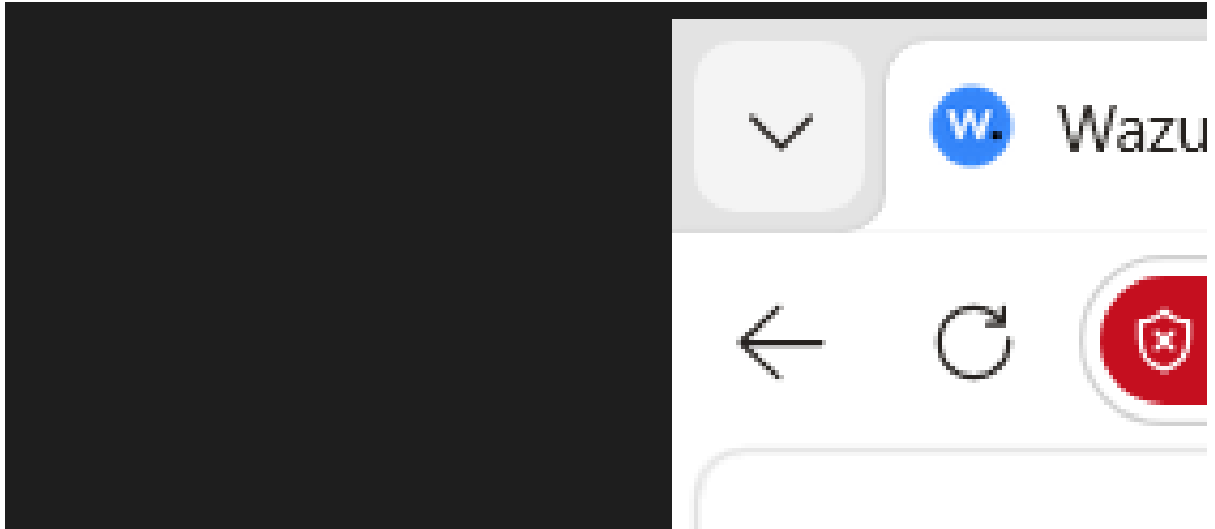


The screenshot displays the Wazuh Threat Hunting dashboard. The browser address bar shows the URL <https://10.0.2.15/app/threat-hunting#/overview/?tab=gen...>. The dashboard header includes the Wazuh logo and a 'Threat Hunting' tab. Below the header, a summary bar indicates '8 hits' for the time range 'Jun 13, 2026 @ 16:22:56.537 - Jun 20, 2026 @ 16:22:56.538'. Action links for 'Export Formatted', 'Reset view', '716 available fields', 'Columns', 'Density', and '1 fields sorted' are present. A table lists the events with columns for timestamp, agent.name, rule.description, rule.level, and rule ID.

timestamp	agent.name	rule.description	rule.level	rule
Jun 20, 2026 @ 14:19:03.5...	WIN-ENDPOINT	Service sta...	3	611
Jun 20, 2026 @ 14:18:15.0...	WIN-ENDPOINT	Service sta...	3	611
Jun 20, 2026 @ 14:13:58.7...	WIN-ENDPOINT	Wazuh age...	3	503
Jun 20, 2026 @ 14:13:46.4...	WIN-ENDPOINT	Wazuh age...	3	506
Jun 20, 2026 @ 14:13:45.9...	WIN-ENDPOINT	Active resp...	3	657
Jun 20, 2026 @ 14:13:39.3...	WIN-ENDPOINT	Summary e...	4	606
Jun 20, 2026 @ 14:13:39.3...	WIN-ENDPOINT	Summary e...	4	606
Jun 20, 2026 @ 14:13:37.6...	WIN-ENDPOINT	New wazu...	3	501

Collected endpoint events and Wazuh agent activity visible in the dashboard.

Wazuh Interface



Wazuh dashboard interface used during monitoring and validation.

Skills Demonstrated

Linux Administration, Windows Administration, SIEM Deployment, Sysmon Integration, Threat Hunting, Endpoint Monitoring, Virtualization, Networking, Troubleshooting, Security Documentation, and Blue-Team Operations.

Future Improvements

Active Directory integration, custom Wazuh detection rules, PowerShell detections, failed login monitoring, MITRE ATT&CK; mapping, and incident response playbooks.